

Enrico Wegner

Contact

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Research Fields

Primary: Macroeconometrics, Time Series Econometrics

Secondary: Causal Inference and Policy Evaluation, Bayesian Econometrics, Macroeconomics

Education

PhD in Econometrics Feb 2023 – exp. 2027

School of Business and Economics, Maastricht University

Advisors: Prof. Stephan Smeekes, Assoc. Prof. Ines Wilms, Asst. Prof. Lenard Lieb.

MSc Economic and Financial Research (specialisation Econometrics) Sep 2020 – Feb 2023

School of Business and Economics, Maastricht University

Graduated Summa Cum Laude (GPA 9.43/10).

BSc Econometrics and Operations Research Sep 2017 – Aug 2020

School of Business and Economics, Maastricht University

Graduated Summa Cum Laude (GPA 9.49/10).

Chapters in PhD Thesis (in progress)

Transmission Channel Analysis in Dynamic Models

Job Market Paper

with Lenard Lieb, Stephan Smeekes, and Ines Wilms

Available as [arXiv:2405.18987](https://arxiv.org/abs/2405.18987).

We propose a framework for the analysis of transmission channels in a large class of dynamic models. To this end, we formulate our approach both using graph theory and potential outcomes, which we show to be equivalent. Our method, labelled Transmission Channel Analysis (TCA), allows for the decomposition of total effects captured by impulse response functions into the effects flowing along transmission channels, thereby providing a quantitative assessment of the strength of various transmission channels. We establish that this requires no additional identification assumptions beyond the identification of the structural shock whose effects the researcher wants to decompose. Additionally, we prove that impulse response functions are sufficient statistics for the computation of transmission effects. We demonstrate the empirical relevance of TCA for policy evaluation by decomposing the effects of policy shocks arising from a variety of popular macroeconomic models.

The Anatomy of Monetary Policy Transmission: Which channels matter?

with Kristina Bluwstein and Michael McLeay (Bank of England)

Under policy embargo; available upon request.

Understanding how monetary policy transmits through the financial system to the real economy and inflation is central to the design, calibration, and evaluation of monetary policy. We study the transmission channels of UK monetary policy using a medium-sized Bayesian VAR identified with an external instrument and apply a novel technique, Transmission Channel Analysis (TCA) based on directed acyclic graphs, to formally define and decompose transmission channels in a dynamic and robust way. Empirically, we find that the bulk of UK monetary policy transmission operates through financial-market and expectations channels. Second-round (feedback) effects become important for output and prices at longer horizons, but are less relevant for consumption. A more granular decomposition shows that housing-related wealth and cash-flow channels account for most of the transmission to real activity and prices; equity wealth and expectations channels are meaningful, while the exchange-rate channel plays a limited role for aggregate consumer prices over the estimation period.

Transmission Channel Analysis without Invertibility

with Tobias Hartl

Work in progress.

We show that transmission channels in the sense of Wegner et al. (2025) can be defined in any linear state space model, including systems with fewer or more shocks than observed variables. This extends transmission channel analysis to non-invertible processes and to all linearised DSGE models, including those with news shocks. We further show that the large directed acyclic graph implied by the state space representation reduces to a much smaller set of variables that remains sufficient to define channels. Reducing the system to a single shock leaves channels contemporaneously indistinguishable in general, but dynamically distinguishable, which permits the analysis of single-shock models and of empirical systems restricted to the shock of interest, at the cost of a change in interpretation. We apply the framework to the forward guidance puzzle in a small-scale New Keynesian model, decomposing the response to a forward guidance shock into inflation expectations, output expectations, and output channels. Inflation expectations dominate under both infinitely lived agents and overlapping generations. Cognitive discounting by households weakens predominantly the output expectations channel, whereas cognitive discounting by firms strengthens that channel and eventually reverses the sign of the inflation expectations channel.

Publications

Esener, S., Wegner, E., Almeida, R. J., Baştürk, N., and Rodrigues, P. (2024). Quantifying Uncertainty of Portfolios using Bayesian Neural Networks. *2024 International Joint Conference on Neural Networks (IJCNN)*. DOI: [10.1109/IJCNN60899.2024.10650382](https://doi.org/10.1109/IJCNN60899.2024.10650382)

Policy and Statistical Reports

Miao, G. and Wegner, E. (2022). Recent Trends in Transport and Insurance Costs and Estimates at Disaggregated Product Level. *OECD Statistics Working Papers No. 2022/02*. DOI: [10.1787/b5dbab02-en](https://doi.org/10.1787/b5dbab02-en)

Miao, G. and Wegner, E. (2022). Using Unit Value Indices as Proxies for International Merchandise Trade Prices. *OECD Statistics Working Papers No. 2022/01*. DOI: [10.1787/27a5abd7-en](https://doi.org/10.1787/27a5abd7-en)

Research and Professional Experience

Bank of England

Monetary Analysis Division

PhD Intern

Jul – Sep 2025

Contributed to the November 2025 Monetary Policy Report, Box B.

Organisation for Economic Co-operation and Development (OECD)

Trade and Productivity Statistics Unit

External Consultant

Feb – Aug 2022

Developed the initial methodology behind the OECD AIS Vessel Tracking Dashboard.

Intern

Jul 2021 – Jan 2022

Updated and evaluated the ITIC and Merchandise Trade Price Index databases.

School of Business and Economics, Maastricht University

Research Assistant (Supervisor: Nalan Baştürk)

Feb – Aug 2022

Bayesian Neural Networks for financial risk management.

Student Assistant (Supervisor: Ines Wilms)

Apr – Jul 2021

Contributed to software and interactive learning material.

Teaching Assistant

Oct 2020 – Apr 2021

Honours and Awards

Selected participant, Lindau Nobel Laureate Meeting in Economic Sciences	2025
Summa Cum Laude, MSc Economic and Financial Research (GPA 9.43/10)	2023
Summa Cum Laude, BSc Econometrics and Operations Research (GPA 9.49/10)	2020

Presentations and Conference Participation

Presentations

Virtual Israel Macro Meeting (invited)	2026
3rd Frankfurt Summer School	
EEA-ESEM Congress	
CEBRA Annual Meeting	
RCEA International Conference	
VTSS Workshop	2025
EC ²	2024
Deutsche Bundesbank Vo-Seminar	
Cornell Statistics Seminar	
Macro, International and Labour Economics Seminar, Maastricht University	2023
The Netherlands Econometric Study Group	2023, 2025
European Seminar on Bayesian Econometrics	2022, 2023
Econometrics Seminar, Maastricht University	2022, 2024

Participation

NBER-NSF Time Series Conference (2024); POLICYMETRICS Summer School, Barcelona School of Economics (2024); The Netherlands Econometric Study Group (2024, 2026); Econometric Methods of Climate Change (2023)

Teaching Experience

Tutor, Maastricht University. Figures are average tutor grades (out of 10) by academic year.

Probability Theory for BSc. Econometrics and Operations Research

Undergraduate tutorial (6.5 ECTS). Introduces the probabilistic foundations of statistics and econometrics, covering random variables and probability distributions, dependence between random variables, and convergence.

Average tutor grade: 9.17 (2025/26); 9.1 (2024/25); 9.27 (2023/24); 9.23 (2022/23).

Macroeconomics for BSc. Econometrics and Operations Research

Undergraduate tutorial (6.5 ECTS). Introduces modern macroeconomic theory and policy, covering short-run fluctuations, medium-run equilibrium, and long-run growth, with applications to European and global economic policy.

Average tutor grade: 9.1 (2025/26); 9.05 (2024/25); 9.15 (2023/24).

Quantitative Methods 3 for BSc. International Business

Undergraduate skills training (4 ECTS). Applies inferential statistical tools, including t-tests, ANOVA, chi-square tests, and regression analysis, to real-world business case studies using SPSS.

Average tutor grade: 8.9 (2025/26); 8.8 (2024/25); 8.35 (2023/24).

Statistics for BSc. Business Analytics / Business Engineering

Undergraduate tutorial (6.5 / 5.0 ECTS). Introduces descriptive statistics, probability theory, and sampling as foundations for statistical inference, including hypothesis testing and regression analysis.

9.73 (Business Analytics, 2020/21); 9.37 (Business Engineering, 2020/21).

Quantitative Business (Statistics) for BSc. Economics and Business Economics (2022/23)

Undergraduate tutorial (6.5 ECTS). Extends first-year quantitative methods, applying mathematical and inferential statistical tools, including regression, ANOVA, and logistic regression, to business economics case studies using SPSS.

Tutors were not individually evaluated due to the course structure.

Thesis Supervision, BSc. and MSc. Econometrics and Operations Research

Topics include Bayesian Neural Networks for financial risk forecasting, econometric and machine-learning methods for Marketing Mix Modelling, and macroeconomic analysis of monetary policy and oil supply shocks.

Software

<i>Creator and Maintainer</i>	TransmissionChannelAnalysis.jl; TCA MATLAB Toolbox; HamiltonFilters.jl; FredMDQD.jl; BayesFlux.jl; BayesFluxR
<i>Contributor</i>	bigtime
<i>Teaching Material</i>	BigTime; VARs; Shell, Git, GitHub

Technical and Language Skills

Programming: C++, Julia, Matlab, Python, R, Stata

Data and Databases: large trade data sets (millions of observations); PySpark pipelines; SQL

Languages: German (native), English (fluent)

Service

<i>Refereeing</i>	Journal of Computational and Graphical Statistics; Empirical Economics; Theory and Decision
<i>Departmental</i>	PhD representative on the strategy review of the Department of Quantitative Economics (2025)
<i>Outreach</i>	Set up and manage the department's LinkedIn presence, jointly with Asst. Prof. Lissa Melis (since 2025); currently redeveloping the departmental website

References

Stephan Smeekes	<i>PhD advisor</i> Professor of Econometrics, Maastricht University s.smeekes@maastrichtuniversity.nl
Ines Wilms	<i>PhD advisor</i> Associate Professor, Maastricht University i.wilms@maastrichtuniversity.nl
Lenard Lieb	<i>PhD advisor</i> Assistant Professor, Maastricht University l.lieb@maastrichtuniversity.nl